

Design and Evaluation of Multi-Agent AI Oracle Systems for Prediction Market Resolution

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1 Introduction

Prediction markets aggregate collective intelligence to forecast uncertain events, but their utility depends on reliable resolution of outcomes. Current oracle systems—the mechanisms that determine whether real-world events occurred—fall into two camps: fully automated oracles like Chainlink that excel at objective data feeds but struggle with subjective outcomes, and human arbitration systems like UMA and Kalshi that achieve high accuracy but incur significant latency and cost. Additionally, human arbitration systems are vulnerable to manipulation by wealthy participants who can stake enough tokens to swing outcomes in their favor. Recent work has demonstrated that large language models can serve as intermediate solutions, with Chainlink’s recently validating a DSPy-based single-LLM oracle on 1,660 Polymarket outcomes and UMA’s OOTruthBot achieving approximately 90% accuracy at roughly \$0.005 per resolution [1] [2].

However, single-LLM systems inherit all failure modes of their underlying model: temporal reasoning errors, hallucinations, and no mechanism for self-correction. A natural question emerges: can multi-LLM architectures improve oracle resolution accuracy over single-model approaches? This thesis investigates whether multi-agent deliberation and aggregation can systematically improve oracle resolution accuracy over single-model baselines, and under what conditions such improvements occur.

We evaluate two multi-agent architectures, independent aggregation and deliberative consensus, for prediction market resolution tasks using KalshiBench, a benchmark of 1500+ questions from Kalshi with verified real-world outcomes [3]. By isolating reasoning from retrieval through a shared evidence layer, we hope to provide controlled empirical evidence on when and why multi-agent approaches succeed or fail. The findings will inform the design of hybrid AI-human oracle systems with principled escalation criteria.

2 Background/Related works

2.1 Oracle Systems for Prediction Markets

Current prediction market oracles span a spectrum from pure automation to full human adjudication. Kalshi resolves markets through a fully manual process in which an internal operations team verifies outcomes against pre-specified, authoritative public sources and records the result [4]. UMA Protocol powers Polymarket through an optimistic oracle where proposers bond assertions that become final if undisputed for 2–48 hours, escalating to token-holder voting when challenged [5]. Kleros takes a different approach through crowdsourced juror panels coordinated by Schelling point incentives [6].

2.2 LLM-Based Oracle Systems

Hybrid human–AI oracle systems are emerging but remain immature. Chainlink’s 2025 research validated a DSPy-based LLM oracle on 1,660 Polymarket outcomes with trading volumes exceeding \$100,000, achieving 99.7% accuracy on sports questions but only 85% on crypto and 84.3% on politics categories. Notably, this study employed a single-LLM architecture and did not filter for potential training data contamination. UMA’s OOTruthBot represents a deployed AI oracle integration, proposing resolutions that achieve over 90% accuracy while reducing costs to approximately \$0.005 per request, with human voters retaining full dispute authority .

2.3 Multi-Agent Consensus Methods

Structured multi-agent approaches have demonstrated improvements over single-model baselines. The Iterative Consensus Ensemble framework loops three LLMs that critique each other until consensus emerges, improving accuracy by 7–15 percentage points over the best single model on medical and general reasoning benchmarks [8]. Most questions settle within 2–3 rounds, keeping computational costs manageable. Furthermore, research on multi-agent debate demonstrated that arithmetic accuracy improves from approximately 70% (single agent) to approximately 95% with three agents over two debate rounds [9].

3 Problem Description

Despite promising results from single-LLM oracle systems, no work has applied multi-agent techniques to prediction market resolution, where correctness directly impacts financial outcomes and system trust. Existing studies on multi-agent debate focus on general reasoning benchmarks, leaving unclear whether reported improvements transfer to the distinct challenges of oracle resolution: interpreting natural-language resolution criteria, verifying time-bounded real-world events, and adjudicating inherently ambiguous outcomes.

Moreover, no prior work has characterized error correlation specifically in oracle settings. While correlated failures across LLMs are known in general reasoning tasks, it remains unknown whether oracle-specific failure modes, such as source verification errors and/or misinterpretation of market conditions, exhibit similar or distinct correlation patterns. Without this understanding, aggregating multiple models may offer little robustness benefit and may instead amplify shared biases.

Finally, current hybrid AI–human oracle systems rely on ad-hoc escalation policies, with no principled framework for determining when model confidence is insufficient and human arbitration justifies its cost and latency. This lack of structure limits the scalability and reliability of hybrid oracle designs.

This thesis addresses these gaps through a controlled empirical evaluation of multi-agent oracle architectures and by developing escalation criteria that balance accuracy, cost, and dispute risk in hybrid AI–human systems.

4 Planned Approaches

The study will use KalshiBench, a benchmark of 1500+ prediction market questions from Kalshi with verifiable real-world outcomes occurring after model training cutoffs. Exa will serve as the shared evidence retrieval layer for all LLM agents, retrieving relevant sources filtered by publication date to include only information that would have existed at the time of resolution. Unlike Chainlink’s evaluation, which did not filter for potential training data contamination, our evidence retrieval is filtered by publication date to ensure models only access information that would have existed at the time of resolution. This shared evidence set will be passed identically to all agents in both architectures, isolating reasoning capability from retrieval capability.

Two multi-agent architectures will be evaluated using GPT-4o, Claude Haiku, and Gemini 2.0. Architecture A (Independent Aggregation) has each model resolve questions independently, with outcomes determined by majority voting and confidence-weighted voting. Architecture B (Deliberative Consensus) has models participate in structured debate: each generates an answer with reasoning, then reviews the other models’ responses and may revise its position before submitting a final answer

Both architectures will be evaluated against single-LLM baselines on accuracy metrics (resolution accuracy with category-wise breakdown) and system behavior metrics (disagreement rate, abstention rate, and convergence patterns). A failure mode taxonomy will analyze 50-100 incorrect resolutions, categorizing errors by type (e.g., temporal reasoning, source verification) and whether they are shared across models or corrected by agent diversity. Based on these findings, the study will develop escalation criteria for hybrid AI–human oracles, defining decision rules based on confidence levels and inter-agent disagreement for routing between automated and human resolution.

5 Schedules / Timelines

This timeline provides a general overview of expected project checkpoints and deadlines:

Jan 29 Project proposal submission & complete literature review

Feb 6 Proposal presentation; complete Exa integration and retrieval pipeline

Feb 13 Implement Architecture A (independent aggregation) with all three models

Feb 20 Implement Architecture B (deliberative consensus)

Feb 27 Run full evaluation on KalshiBench for both architectures

Mar 6 Midterm progress report due; complete accuracy analysis

Mar 13–20 Spring recess; begin failure mode analysis

Mar 27 Complete failure mode taxonomy (50–100 cases)

Apr 3 Midterm presentation; develop escalation framework

Apr 10 Evaluate escalation criteria; cost–accuracy tradeoff analysis

Apr 17 Complete writing of results and discussion

Apr 20 Final report draft due to advisor

Apr 22 Final poster submission

Apr 30 Final report submission

6 Deliverables

This project will produce an empirical analysis comparing independent aggregation and deliberative consensus architectures against single-LLM baselines, along with a failure mode taxonomy characterizing when errors are shared across models versus corrected by agent diversity. The study will also develop an escalation framework with decision rules for routing between AI and human resolution in hybrid systems. All code implementing the multi-agent architectures with Exa integration will be released as a public GitHub repository, accompanied by a final report and poster documenting methodology, results, and implications for oracle system design.

References

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